

Semantic representations : word embeddings and named entity recognition

Lab report, NLP Today course

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1 Introduction

This report presents our work for the lab sessions of the Semantic Representations part of the course. The first lab consists in building and comparing static word embeddings, that is representations where every occurrence of a word gets the same vector whatever its context. We use three approaches: word2vec CBOW, word2vec skip-gram and fastText CBOW. Each approach is trained on two French corpora that differ in size and in domain, a small medical corpus (QUAERO_FrenchMed) and a larger news corpus (QUAERO_FrenchPress), which gives six models.

Following the instructions, we compare these models through semantic similarity: for five candidate words (*patient*, *traitement*, *maladie*, *solution* and *jaune*), we look at the ten closest words in terms of cosine similarity. This evaluation relies on the idea stated in the course: if two word vectors are close, the words should be semantically or syntactically close. The embeddings will then be reused in the second lab for named entity recognition.

2 Lab 1: word embeddings training

2.1 Training setup

Corpora. Both files are already tokenized, with one sentence per line and tokens separated by spaces. Our preprocessing is lowercasing, so that forms such as *Traitement* and *traitement* are counted as the same word, which matters when the corpus is small. Table 1 gives the main figures.

Corpus	Sentences	Tokens	Vocabulary
QUAERO_FrenchMed	3,021	51,896	8,088
QUAERO_FrenchPress	38,548	1,251,561	38,279

Table 1: Size of the two training corpora after lowercasing

The news corpus contains about 24 times more tokens than the medical one. This difference is important for the comparisons below.

Approaches. The three approaches learn the embeddings while solving a word prediction task:

- CBOW predicts the current word from the average of the embeddings of its context words;
- skip-gram does the opposite: from the current word, it predicts each word of its context;
- fastText represents a word by the average of its own vector and of the vectors of its character n-grams. We use its CBOW version, as asked in the instructions. Thanks to the n-grams, fastText can also build a vector for a word that never appears in the training data.

Hyperparameters. As recommended in the instructions, we use vectors of dimension 100 and `min_count = 1`, so that every word of the corpus gets a vector. The context window is 5 (up to five words on each side of the current word) and the other parameters keep their gensim default values (5 training epochs, negative sampling with 5 negative words). The six models are trained with the `Word2Vec` and `FastText` classes of the gensim library.

To make the results reproducible, we fixed the random seed (`seed = 42`) and trained on a single thread (`workers = 1`). With several threads, the order of the updates changes from one run to another, and the nearest neighbours change slightly even with a fixed seed. Finally, the vectors of the six models are saved in the word2vec binary format, to be reused in the second lab.

2.2 Semantic similarity: same corpus, different approaches

To measure the impact of the approach, we compare the three models trained on the medical corpus. Table 2 gives the first five neighbours of each candidate word (the full lists of ten neighbours are in the notebook) and, in the last row, the range of the cosine similarities of the ten neighbours.

Word	CBOW	Skip-gram	fastText
patient	par, en, avec, pendant, dans	qui, symptômes, recommandé, devra, nécessaire	patiente, conscients, ingrédients, présentaient, pansements
traitement	le, par, des, sur, dans	tysabri, le, médicaments, tasmart, ayant	taaitement, allaitement, évitement, traitements, étroitement
maladie	en, des, la, avec, du	association, la, acide, administration, infection	malade, prolactine, sociale, héparine, de
solution	contient, flacon, (, mg,)	flacon, contient, 100, mg, jour	dissolution, évolution, evolution, dilution, pollution
jaune	ou, avec, pour, ces, résultats	13, puis, 24, boîtes, ±	une, immune, minérale, intestinale, anale
Cosine range	0.998 to 1.000	0.945 to 0.998	above 0.999

Table 2: First five nearest neighbours of the candidate words for the three models trained on the medical corpus

CBOW. The CBOW model does not produce usable neighbours. For four words out of five, the closest words are function words (*le, par, des, en*) or punctuation, and all cosine similarities lie between 0.998 and 1.000. When every neighbour gets almost the same score, the ranking no longer means anything: the candidate words and their neighbours have almost the same direction, and the most frequent words come first because they appear in every context. Only *solution* gets related words (*contient, flacon, mg*), because it appears very often in the same fixed expressions of drug leaflets (*un ml de solution contient ...*).

We explain this failure by the small size of the corpus combined with the way CBOW learns. In CBOW, each position of the text gives a single prediction, made from the merged context, whereas

skip-gram turns every occurrence of a word into several training examples, one per context word. With about 52,000 tokens and 5 epochs, CBOW does not receive enough training signal to separate the words. This agrees with the authors of word2vec, who present skip-gram as better suited to infrequent words.

Skip-gram. With skip-gram, the neighbours start to make sense in the medical domain. *traitement* is close to *médicaments* and to drug names (*tysabri*, *tasmar*, and *epivir* further in the list), that is, to actual treatments. *solution* is close to *flacon*, *mg* and *injectable*: the model has learned the pharmaceutical meaning of the word (a liquid form of a drug, as in *solution pour perfusion*), not its everyday meaning. *maladie* gets *infection* and *cancer* among its ten neighbours, and *patient* gets *symptômes* and *administré*. The scores are also lower and more spread (0.945 to 0.998), which makes the ranking more informative.

Two limits remain. Some function words are still present (*le*, *qui*, *la*), and *jaune*, which occurs only 9 times, gets numbers and packaging words (*boîtes*, and *titane* further in the list). The corpus explains these neighbours: *jaune* mostly appears in descriptions of tablets and in lists of excipients (*oxyde de fer jaune*, *dioxyde de titane*). The model therefore reflects the use of the word in this corpus, but 9 occurrences are not enough to get a reliable representation.

fastText. The fastText neighbours are of a different nature: they share long character sequences with the candidate word. *patient* is close to *patiente*, *maladie* to *malade*, and *traitement* to *traitements* and even to the misspelled forms *taaitement* and *traitment*, which occur in the corpus. This is the direct effect of the character n-grams: two words that share many n-grams get close vectors. This property is useful, since it groups inflected forms and spelling errors and gives vectors to unseen words, which should help for named entity recognition in the second lab.

However, the similarity then follows the form more than the meaning. *allaitement*, *évitement* and *étroitement* are close to *traitement* only because of the ending *-ement*, *solution* is close to *évolution* and *pollution* because of *-lution*, and *jaune* to *une* and *immune*. Like CBOW, whose architecture it uses here, fastText gives scores above 0.999, which suggests that on such a small corpus the shared n-grams weigh more than the contexts in the final vectors.

2.3 Semantic similarity: same approach, different corpora

We now compare each approach across the two corpora to measure the impact of the data. Two factors change at the same time: the quantity of data (24 times more tokens in the news corpus) and its type (news instead of medical texts). Table 3 gives the first five neighbours for the models trained on the news corpus, together with the number of occurrences of each candidate word in both corpora.

Word	Occurrences	CBOW	Skip-gram	fastText
patient	34 / 11	artificiel, morceau, délit, constituer, éthique	motivation, accumulé, identification, brevet, accuser	patientent, impatient, trient, détient, coefficient
traitement	288 / 53	contenu, trésor, collectif, fréjus, nucléaire	mépris, trésor, potentiel, coût, fonctionnement	concrètement, comportement, farouchement, promptement, fondement
maladie	72 / 114	reprise, somme, douleur, bataille, négociation	prime, drogue, décote, pauvreté, nationalité	malnutrie, trilogie, magie, ossétie, démographie
solution	70 / 100	seule, difficulté, querelle, garantie, suffisante	légitimité, pacifique, consensuelle, difficulté, crédibilité	révolution, résolution, évolution, pollution, dissolution
jaune	9 / 23	michael, baba, peter, maréchal, lars	maillot, cooke, emparé, sprint, baden	abdoune, lune, brune, saadoune, saâdoune
Cosine range		0.855 to 0.949	0.814 to 0.954	0.921 to 0.992

Table 3: First five nearest neighbours of the candidate words for the three models trained on the news corpus. The column (Occurrences) gives the number of occurrences of each word in the medical and in the news corpus

Quantity of data. With 24 times more tokens, the similarity scores are no longer saturated: they range from 0.855 to 0.949 for CBOW and from 0.814 to 0.954 for skip-gram, against 0.998 to 1.000 and 0.945 to 0.998 on the medical corpus. The vectors are therefore better spread in the space, and the ranking of the neighbours becomes meaningful. The most visible change concerns CBOW: its neighbours are no longer function words but content words (*difficulté* and *garantie* for *solution*, *douleur* for *maladie*). The failure observed in the previous section was therefore mainly due to the lack of data. For fastText, on the contrary, the larger corpus does not change the nature of the neighbours, which still share the ending of the candidate word.

Type of data. More data does not mean better neighbours for our candidate words, because the news corpus does not use them in the same way. Most of them are medical words, and the news corpus uses them either less often (*patient* appears 11 times against 34, *traitement* 53 times against 288) or with another meaning. Looking at the corpus explains the neighbours given by skip-gram :

- *solution* is used in its everyday meaning (*trouver une solution, une solution de compromis*), hence neighbours such as *consensuelle*, *pacifique* or *alternative*, far from the pharmaceutical meaning learned on the medical corpus;
- *traitement* appears in social and economic contexts (*le traitement du chômage, la différence de traitement entre les salariés*), hence *coût*, *financement* or *financier*;
- *maladie* often appears in social debates (*assurance maladie* occurs 19 times), hence *prime*, *décote* or *pauvreté*;
- *jaune* is mostly used in sports news: 14 of its 23 occurrences are in *maillot jaune*, which explains *maillot* and *sprint*;
- *patient*, with only 11 occurrences, some of them as an adjective (*être patient*), gets neighbours without a clear link to it.

3 Conclusion

This first lab shows that static word embeddings depend on both the training approach and the training data. On the small medical corpus, skip-gram gives the most meaningful neighbours, CBOW does not manage to separate the words (all its similarity scores are close to 1), and fastText mainly captures the form of the words, which groups inflected forms and spelling errors. With 24 times more data, the news corpus gives a better structured space, where CBOW finally works, but the meanings learned are those of the news domain: *solution* becomes the solution to a problem, *traitement* gets a social and economic meaning, and *jaune* is associated with the *maillot jaune*. In short, the quantity of data decides whether a model manages to learn, and the type of data decides what it learns.

This evaluation remains qualitative and relies on only five words, some of which are rare (*jaune* occurs 9 times in the medical corpus), so these conclusions need to be confirmed on a real task.